Supplement to "Linnik's constant below the zero-density exponent: 11/5 from a zero-free half-plane and the Chen–Gupta–Li density estimate"

Ritchie Comley

Draft, 30 October 2026

This supplement holds the proofs of four standard results that the paper uses, of its kernel bound, and of Lemma 4.3 and Proposition 4.8, in the form in which the Lean library proves them (Sections S1 to S4), and the details of the paper's account of the formalisation (Section S5) and of its computations around the exponent (Section S6). Numbers such as Lemma 3.2 and Section 5 refer to the paper, and those that begin with S to this supplement. Its notation is the paper's.

S1 Proof of Lemma 3.2, and the kernel bound

Proof. Throughout, $Q \geq 4$, so that $\log Q > 1$, and L_χ stands for $L(\cdot, \chi)$, which is entire since χ is not trivial. Put $\sigma_* = 11/12 + \delta$ and $\tau = (1+H)^{-J}$.

The weight. Let $f_Q(u) = f_0(u) \log Q / \log(uQ)$ where $f_0(u) \neq 0$, which needs $u \geq 1/2$ and so $uQ \geq 2$, and $f_Q(u) = 0$ elsewhere; and let $w_y(u) = u^{-iy} f_Q(u)$. If $f_0(n/Q) \neq 0$ then $n \geq Q/2 \geq 2$ and $f_Q(n/Q) = f_0(n/Q) \log Q / \log n$, so that $\Lambda(n) f_Q(n/Q) = \log Q \cdot (\Lambda(n)/\log n) f_0(n/Q)$. Since $\Lambda(p)/\log p = 1$,

$$\sum_n \chi(n)\Lambda(n) w_y(n/Q) = \log Q \, \big(S_Q(\chi, y) + R\big),$$

$$R = \sum_{\substack{p^k \leq 2Q \\ k \geq 2}} \frac{\chi(p^k)}{k} \Big(\frac{p^k}{Q}\Big)^{-iy} f_0\Big(\frac{p^k}{Q}\Big).$$

Each term of R is at most $C = \max|f_0|$ in absolute value, and a proper prime power $p^k \le 2Q$ has $p \le \sqrt{2Q}$ and $k \le \log_2(2Q)$, so $|R| \le C(\sqrt{2Q}+1)(\log_2(2Q)+1)$. It remains to prove

$$\Big|\sum_n \chi(n)\Lambda(n)w_y(n/Q)\Big| \le M_\delta(\theta)\,\lambda\,B\big(Q^{\theta+\delta}+Q^{\sigma_*}\tau\big). \tag{S1.1}$$

The Mellin transform. Write $\mathcal{M}f(s) = \int_0^\infty u^{s-1}f(u)\,du$. The substitution $u = e^{-v}$ gives, with $c_Q = 1/\log Q \in (0, 1/\log 4]$,

$$\mathcal{M}f_Q(\sigma+it) = \int_{\mathbb{R}} e^{-itv}V(v)\,dv, \qquad V(v) = e^{-\sigma v}f_0(e^{-v})(1-c_Q v)^{-1}.$$

The function V vanishes for $|v| > \log 2$, and $c_Q v \le 1/2$ for $v \le \log 2$; the pole $v = \log Q \ge \log 4$ of $(1-c_Q v)^{-1}$ lies where V vanishes, so V is smooth. By Leibniz's rule, with $|(d/dv)^i(1-c_Q v)^{-1}| = i!\,c_Q^i|1-c_Q v|^{-1-i} \le i!\,2^{i+1}$ on $|v| \le \log 2$, the j-th derivative of V is bounded by a constant V_j that depends only on f_0 and j, uniformly in $0 \le \sigma \le 2$ and $Q \ge 4$. Integration by parts j times gives $|t|^j|\mathcal{M}f_Q(\sigma+it)| \le 2\log 2 \cdot V_j$, and with $j = 0$ and $j = J+3$,

$$(1+|t|)^{J+3}|\mathcal{M}f_Q(\sigma+it)| \le B \qquad (0 \le \sigma \le 2,\ t \in \mathbb{R},\ Q \ge 4),$$

where $B = 2^{J+4}\log 2\,(V_0 + V_{J+3})$ depends only on f_0 and J. Since f_Q is continuous on $(0,\infty)$ and vanishes near 0 and near ∞, $\mathcal{M}f_Q$ is entire. Finally $\mathcal{M}w_y(s) = \mathcal{M}f_Q(s-iy)$, so that

$$(1+|t-y|)^{J+3}|\mathcal{M}w_y(\sigma+it)| \le B \qquad (0 \le \sigma \le 2,\ t \in \mathbb{R}). \tag{S1.2}$$

Perron's formula. For $\operatorname{Re} s > 1$, $-L'_\chi/L_\chi(s) = \sum_n \chi(n)\Lambda(n)n^{-s}$. Mellin inversion on the line $\operatorname{Re} s = 2$ at each point n/Q, with the exchange of sum and integral justified by (S1.2), gives

$$\sum_n \chi(n)\Lambda(n)w_y(n/Q) = \frac{1}{2\pi}\int_{\mathbb{R}} F(2+it)\,dt,$$

$$F(s) = -\frac{L'_\chi}{L_\chi}(s)\,\mathcal{M}w_y(s)\,Q^s.$$

The logarithmic derivative. For $1/2 \le \theta_0 < \theta' < 3/2$ there is $C_{\log}(\theta_0,\theta')$ with the following property, for every q, every non-trivial $\chi \bmod q$ and every $t \in \mathbb{R}$: if L_χ has no zero in the open disc D_t of centre $3/2+it$ and radius $\rho = 3/2-\theta_0$, then

$$\Big|\frac{L'_\chi}{L_\chi}(\sigma+it)\Big| \le C_{\log}(\theta_0,\theta')\big(\log(q+2)+\log(|t|+4)\big) \qquad (\sigma \ge \theta'). \tag{S1.3}$$

For $\sigma \ge 3/2$ this holds as soon as $C_{\log}(\theta_0,\theta') \ge -\zeta'/\zeta(3/2)$, by the Dirichlet series of L'_χ/L_χ. For $\theta' \le \sigma < 3/2$ it follows from a lemma of Landau's type: if f is analytic near the closed unit disc, $f(0)=1$, $|f| \le B_0$ on $|z| \le R$ with $B_0 > 1$, f has no zero in $|z|<1$ and $0 < r' < r < R' < R < 1$, then

$$|f'/f(z)| \le \Big(\frac{16r^2}{(r-r')^3} + \frac{1}{(R^2/R' - R')\log(R/R')}\Big)\log B_0 \qquad (|z| \le r').$$

(The library uses the form of this lemma in PrimeNumberTheoremAnd [51], `FinalBound`, which combines a Blaschke product, the Borel–Carathéodory inequality and Jensen's formula, and allows zeros.) Apply it to $f(z) = L_\chi(3/2+it+\rho z)/L_\chi(3/2+it)$ at $z = (\sigma-3/2)/\rho$, with $r' = (3/2-\theta')/\rho$ and with r, R' and R a quarter, a half and three quarters of the way from r' to 1; then $f'/f(z) = \rho\, L'_\chi/L_\chi(\sigma+it)$. The bound B_0 comes from two estimates. First, $|L_\chi(s)| \le 3(q+1)(|\operatorname{Im} s|+2)$ for $1/2 \le \operatorname{Re} s \le 3$, which follows from the partial-summation bound $|L_\chi(s)| \le (q+1)|s|/\operatorname{Re} s$ for $\operatorname{Re} s > 0$. Second, $|L_\chi(3/2+it)|^{-1} \le \zeta(3/2)$, from $1/L_\chi(s) = \sum \mu(n)\chi(n)n^{-s}$. Together they give $|f| \le B_0 = 2 + 6(q+1)\zeta(3/2)(|t|+4)$ on $|z| \le R$, and $\log B_0 \ll \log(q+2) + \log(|t|+4)$. The constant $C_{\log}(\theta_0,\theta')$ is explicit and of order $(\theta'-\theta_0)^{-3}$.

The disc D_t lies in $\operatorname{Re} s > \theta_0$ and, since $\rho \le 1$, in $|\operatorname{Im} s - t| < 1$. So (S1.3) applies in two cases:

(i) with $\theta_0 = 11/12$ and every t, by Input 1;

(ii) with $\theta_0 = \theta$ and every t with $|t-y| \le H$, by the hypothesis of the lemma.

Since $|t|+4 \le (|y|+4)(1+|t-y|)$ and $\log(1+u) \le u$, we have $\log(q+2) + \log(|t|+4) \le \lambda(1+|t-y|)$. With (S1.2) this gives, in either case,

$$|F(\sigma+it)| \le C_{\log}(\theta_0,\theta')\,\lambda\, B\, Q^\sigma (1+|t-y|)^{-J-2} \qquad (\theta' \le \sigma \le 2). \quad \text{(S1.4)}$$

The contour. By Input 1, F is holomorphic on $\sigma_* \le \operatorname{Re} s \le 2$, and by (S1.4) in case (i), with $\theta' = \sigma_*$, it is $O((1+|t-y|)^{-2})$ there uniformly. So Cauchy's theorem on the rectangles $[\sigma_*, 2] \times [-t_0, t_0]$, with $t_0 \to \infty$, moves the line of integration from 2 to σ_*. Split the line $\operatorname{Re} s = \sigma_*$ at the heights $y \pm H$. By the hypothesis of the lemma, F is holomorphic on the rectangle $\theta+\delta \le \operatorname{Re} s \le \sigma_*$, $|\operatorname{Im} s - y| \le H$, so by Cauchy's theorem the integral over $|t-y| \le H$ on the line σ_* equals the integral over the same heights on the line $\theta+\delta$ plus the integrals over the two horizontal sides at the heights $y \pm H$.

The pieces. Put $C_1 = C_{\log}(11/12, \sigma_*)$ and $C_2 = C_{\log}(\theta, \theta+\delta)$.

- On the line σ_* with $|t-y| \geq H$: by (S1.4) in case (i) and $(1+u)^{-J-2} \leq (1+H)^{-J}(1+u^2)^{-1}$ for $u \geq H$, $|F| \leq C_1 \lambda B Q^{\sigma_*}\tau/(1+(t-y)^2)$. Each of the two tails contributes at most $\pi C_1 \lambda B Q^{\sigma_*}\tau$.

- On the line $\theta+\delta$ with $|t-y| \leq H$: by (S1.4) in case (ii), $|F| \leq C_2 \lambda B Q^{\theta+\delta}/(1+(t-y)^2)$, so this part contributes at most $\pi C_2 \lambda B Q^{\theta+\delta}$.

- On each horizontal side, of length $11/12-\theta \leq 1$: by (S1.4) in case (ii), with $Q^{\sigma} \leq Q^{\sigma_*}$, $|F| \leq C_2 \lambda B Q^{\sigma_*}(1+H)^{-J-2} \leq C_2 \lambda B Q^{\sigma_*}\tau$.

Adding the pieces and dividing by 2π, the sum in (S1.1) is at most

$$C_1 \lambda B Q^{\sigma_*}\tau + \tfrac{1}{2} C_2 \lambda B Q^{\theta+\delta} + \tfrac{1}{\pi} C_2 \lambda B Q^{\sigma_*}\tau \leq (C_1 + C_2)\, \lambda\, B\big(Q^{\theta+\delta} + Q^{\sigma_*}\tau\big),$$

which is (S1.1) with $M_\delta(\theta) = C_{\log}(\theta, \theta+\delta) + C_{\log}(11/12, 11/12+\delta)$. □

The kernel bound of Section 3.3

Let $\delta' = \varepsilon' = 1/1000$, let ψ be a non-trivial character mod q, and let $Y \geq 1$. By Mellin inversion, $k_Y(\psi) = (2\pi i)^{-1} \int_{(2)} L(s,\psi)\Gamma(s) Y^s\, ds$. Since $L(\cdot,\psi)$ is entire and $1/12-\delta' > 0$, the line moves to $\operatorname{Re} s = 1/12-\delta'$ without crossing a pole. There the functional equation of the primitive character ψ^* that induces ψ, of conductor q^*, gives $|L(s,\psi^*)| \ll (q^*(|\operatorname{Im} s|+2))^{5/12+\delta'} |L(1-s, \overline{\psi^*})|$, and $1-s$ lies on $\operatorname{Re} s = 11/12+\delta'$. There Input 1 gives the Lindelöf-type bound $|L(s,\chi)|^{\pm 1} \ll ((q+2)(|\operatorname{Im} s|+4))^{\varepsilon'}$, uniformly in q and $\chi \neq \chi_0$, by the argument by which the Riemann hypothesis implies the Lindelöf hypothesis [20, §14.2]. The lemma of Landau's type above bounds $|L'/L(s,\chi)|$ by a multiple of $\log((q+2)(|\operatorname{Im} s|+4))$ on $\operatorname{Re} s \geq 11/12+\delta'/2$; the three-lines theorem, on rectangles of height comparable to that logarithm, interpolates this against $|L'/L| \ll 1$ on $\operatorname{Re} s = 3/2$; and integrating L'/L from $\operatorname{Re} s = 3/2$, where $|\log|L|| \leq \log \zeta(3/2)$, gives the bound for $|L|$ and $1/|L|$. The Euler factors at the primes dividing q are at most 2 each, which is $\ll q^{\varepsilon'}$ in all, and together $|L(s,\psi)| \ll (q(|\operatorname{Im} s|+2))^{5/12+\delta'+2\varepsilon'}$ on the new line. The decay of Γ makes the integral converge, and $|k_Y(\psi)| \leq C Y^{1/12-\delta'}(2q)^{5/12+\delta'+2\varepsilon'}$ with C independent of q, ψ and Y.

S2 Proof of Lemma 4.4

Proof. Let the family have the scale P, the set S, the coefficients a_m, the primes $p_1, \ldots, p_K$ at the scales Q_i, and the comparisons $C_1, \ldots, C_J$ with forms ℓ_j and pivots $\operatorname{piv}(j)$. Let $\mathcal{T}$ be the set of the $t = (m, p_1, \ldots, p_K)$ with

$m \in S$ and each p_i a prime at most $2Q_i$; for $t \in \mathcal{T}$ put $w(t) = a_m \prod_i f_0(p_i/Q_i)$ and $n(t) = mp_1 \cdots p_K$, and for a function v on $\mathcal{T}$ put

$$\mathcal{D}(v) = \sum_{t \in \mathcal{T}} v(t)\, \Delta(n(t)), \qquad \Delta(n) = \mathbf{1}[n \equiv a] - \frac{\mathbf{1}[(n,q)=1]}{\varphi(q)}.$$

The sum of the lemma is $\mathcal{D}(w \prod_j I_j)$, with $I_j(t) = \mathbf{1}[C_j \text{ holds at } t]$. Let $F_0 = \max f_0$ and $\eta = 10^{-5}$, the largest height exponent that Theorem 3.1 allows.

Orthogonality. For a prime to q, $\Delta(n) = \varphi(q)^{-1} \sum_{\chi \in \mathcal{X}^*} \overline{\chi}(a)\chi(n)$, so for every v

$$|\mathcal{D}(v)| \le \frac{1}{\varphi(q)} \sum_{\chi \in \mathcal{X}^*} \Big| \sum_{t \in \mathcal{T}} v(t)\chi(n(t)) \Big|. \tag{S2.1}$$

A smooth family. Let $g_1, \dots, g_J$ be continuous and integrable, with integrable Fourier transforms $\hat{g}_j(\xi) = \int g_j(u) e^{-2\pi i u\xi}\, du$, and let $T_0 \ge 1$ with $2\pi \sum_j |e_{ji}|\, T_0 \le q^\eta$ for every i. We show that for q larger than the q_0 of Theorem 3.1 (with f_0, K_0, e_1 and η),

$$\sum_{\chi \in \mathcal{X}^*} \Big| \sum_{t \in \mathcal{T}} w(t) \prod_j g_j(\ell_j(t))\, \chi(n(t)) \Big| \le q^{L - 1/50000} \mathcal{N} \prod_j \|\hat{g}_j\|_1 + \varphi(q)\, \mathfrak{B} \sum_j \int_{|\xi| > T_0} |\hat{g}_j| \prod_{l \ne j} \|\hat{g}_l\|_1, \tag{S2.2}$$

where

$$\mathcal{N} = \Big(\frac{1}{P} \sum_{m \in S} |a_m|^2 \Big)^{1/2} \le A_0, \qquad \mathfrak{B} = \sum_m |a_m| \prod_i \sum_{p \le 2Q_i} f_0(p/Q_i).$$

By Fourier inversion, and since the phase factorises over the variables,

$$\prod_j g_j(\ell_j) = \int_{\mathbb{R}^J} \prod_j \hat{g}_j(\xi_j)\, e^{2\pi i \sum_j \xi_j \ell_j}\, d\xi,$$
$$e^{2\pi i \sum_j \xi_j \ell_j(t)} = e^{2\pi i \sum_j \xi_j \lambda_j(m)} \prod_i p_i^{2\pi i H_i}, \qquad H_i = \sum_j \xi_j e_{ji}.$$

So the inner sum in (S2.2) is $\int \prod_j \hat{g}_j(\xi_j)\, W_\xi(\chi) \prod_i Q_i^{2\pi i H_i} S_{Q_i}(\chi, h_i)\, d\xi$, where $W_\xi(\chi) = \sum_m a_m e^{2\pi i \sum_j \xi_j \lambda_j(m)} \chi(m)$ and $h_i = -2\pi H_i$. Take absolute values, sum over χ, and bound the integrand for each ξ. If $|\xi_j| \le T_0$ for every j, then $|h_i| \le q^\eta$, and Theorem 3.1, with the coefficients $a_m e^{2\pi i \sum_j \xi_j \lambda_j(m)}$, which

have the moduli $|a_m|$, and the heights h_i, gives $\sum_\chi |W_\xi(\chi)| \prod_i |S_{Q_i}(\chi, h_i)| \le q^{L-1/50000}\mathcal{N}$. Otherwise some $|\xi_j| > T_0$, and the trivial bound $\varphi(q)\mathfrak{B}$ holds. Integrating gives (S2.2).

The cut-off. Let g be smooth, with $g = 0$ on $(-\infty, 0]$, $g = 1$ on $[1, \infty)$ and $0 \le g \le 1$, and let $c_k = \int |g^{(k)}|$. For $0 < \nu \le 2\Lambda$ put $G_{\nu,\Lambda}(u) = g(u/\nu) - g((u - 2\Lambda)/\nu)$. It is smooth, $0 \le G_{\nu,\Lambda} \le 1$, it vanishes outside $[0, 2\Lambda + \nu]$, and $G_{\nu,\Lambda}(u) = g(u/\nu)$ for $u \le 2\Lambda$. So if $u \le \Lambda$ and a condition holds when $u > 0$ and fails when $u < 0$,

$$\big|\mathbf{1}[\text{condition}] - G_{\nu,\Lambda}(u)\big| \le G_{\nu,\nu}(u + \nu), \tag{S2.3}$$

since the left side vanishes unless $0 \le u < \nu$, where $G_{\nu,\nu}(u + \nu) = 1$. Integration by parts gives $(2\pi|\xi|)^k |\hat{G}_{\nu,\Lambda}(\xi)| \le \|G^{(k)}_{\nu,\Lambda}\|_1 \le 2\nu^{1-k} c_k$, and $|\hat{G}_{\nu,\Lambda}| \le 2\Lambda + \nu$. Hence

$$\|\hat{G}_{\nu,\Lambda}\|_1 \le \pi(2\Lambda + \nu) + \frac{c_2}{2\pi\nu},$$
$$\int_{|\xi|>T_0} |\hat{G}_{\nu,\Lambda}| \le \frac{4\pi c_k}{(2\pi)^k}\, \nu^{1-k} T_0^{2-k} \qquad (k \ge 2,\ T_0 \ge 1).$$

Sharp against smooth. Let $\Lambda = \Lambda_0 \log q + 1$, so that $\ell_j \le \Lambda$ on $\mathcal{T}$, and put $G_j = G_{\nu,\Lambda}(\ell_j)$ and $M_j = G_{\nu,\nu}(\ell_j + \nu)$. All these values lie in $[0, 1]$, so by (S2.3) $|\prod_j I_j - \prod_j G_j| \le \sum_j |I_j - G_j| \le \sum_j M_j$. Since $|\Delta(n)| \le \Delta(n) + 2 \cdot \mathbf{1}[(n, q) = 1]/\varphi(q)$ and $f_0 \ge 0$,

$$\Big|\mathcal{D}\Big(w \prod_j I_j\Big)\Big| \le \Big|\mathcal{D}\Big(w \prod_j G_j\Big)\Big| + \sum_j \Big(|\mathcal{D}(|w|M_j)| + \frac{2}{\varphi(q)} \sum_{t\in\mathcal{T}} |w(t)| M_j(t)\Big). \tag{S2.4}$$

The first term and each $\mathcal{D}(|w|M_j)$ are smooth families: the first with $g_j = G_{\nu,\Lambda}$, the second with the coefficients $|a_m|$ and the single function $G_{\nu,\nu}$ at the form $\ell_j + \nu$. Both are bounded by (S2.1) and (S2.2). A trivial count would not do for them, since it would lose the factor $1/\varphi(q)$ that the condition $n \equiv a$ gives. The last sum, which carries that factor already, is counted trivially, by the pivot. Fix m and the primes other than $p_{\mathrm{piv}(j)}$; then $\ell_j = c \pm \log p_{\mathrm{piv}(j)}$, and $M_j \ne 0$ forces $-\nu \le \ell_j \le 2\nu$, which confines $p_{\mathrm{piv}(j)}$ to an interval $[Y, e^{3\nu}Y]$. At most $(e^{3\nu} - 1)2Q_{\mathrm{piv}(j)} + 1$ integers up to $2Q_{\mathrm{piv}(j)}$ lie in it, so

$$\sum_{t\in\mathcal{T}} |w(t)| M_j(t) \le F_0\big((e^{3\nu} - 1)2Q_{\mathrm{piv}(j)} + 1\big) \sum_m |a_m| \prod_{i\ne \mathrm{piv}(j)} \sum_{p \le 2Q_i} f_0(p/Q_i).$$

The parameters. Put $\varepsilon_2 = 1/(400000(J_0 + 1))$, $\nu = q^{-\varepsilon_2}$, $T_0 = q^{1/200000}$, $k = 1200003$ and $\delta = \min(\varepsilon_2, e_1)/2$. For large q:

- $2\pi J_0 E_0 T_0 \le q^{\eta}$, so (S2.2) applies, as $\sum_j |e_{ji}| \le J_0 E_0$;
- $\pi(2\Lambda+1) \le q^{\varepsilon_2}$, so $\|\hat{G}_{\nu,\Lambda}\|_1$ and $\|\hat{G}_{\nu,\nu}\|_1$ are at most Cq^{ε_2} with $C = 1 + c_2/(2\pi)$, and a product of at most J_0 of them is at most $C^{J_0+1} q^{1/400000}$;
- $\nu^{1-k} T_0^{2-k} = q^{\varepsilon_2(k-1)-(k-2)/200000} \le q^{-3}$, which bounds the tails.

The constants come from $\eta = 10^{-5}$: the norms lose at most $q^{\eta/4}$, $T_0 = q^{\eta/2}$ leaves room for the factor $2\pi J_0 E_0$ below q^{η}, and $k = 1200003$ is the least k with $(k-1)\eta/4 - (k-2)\eta/2 \le -3$, so that the tails are at most q^{-3} whatever J_0. Moreover $\mathfrak{B} \le P A_0 (3F_0)^K \prod_i Q_i \le A_0 (3F_0+1)^{K_0} 2^{K_0} q^L$, since at most $3Q$ integers lie in $[1, 2Q]$ for $Q \ge 1$ and $P \prod_i Q_i \le 2^{K_0} x$. Finally $e^{3\nu} - 1 \le 6\nu$ for $\nu \le 1/3$ and $Q_{\mathrm{piv}(j)} \ge q^{e_1}$ give $(12\nu Q_{\mathrm{piv}(j)} + 1) \prod_{i \ne \mathrm{piv}(j)} Q_i \le (12\nu + q^{-e_1}) \prod_i Q_i$. Putting these into (S2.4), with (S2.1) and (S2.2), and using $\varphi(q) \le q$, the sum of the lemma is at most

$$\frac{A_0 q^L}{\varphi(q)} \big(a_1 q^{-7/400000} + a_2 q^{-1} + a_3 (12 q^{-\varepsilon_2} + q^{-e_1}) \big),$$

where a_1, a_2 and a_3 depend only on K_0, J_0 and F_0. The three terms come from the smooth families, from the tails and from the pivot counts. Each power of q in the bracket is at most $q^{-2\delta}$, so for large q the sum is at most $A_0 q^{L-\delta}/\varphi(q)$. Here δ depends only on J_0 and e_1, and Theorem 3.1 was used once for the smoothed family and once for each near-boundary family. □

S3 Proofs of Propositions 4.7 and 4.8, and of Lemma 4.3

Proof. Write $\mathcal{P}[y, z)$ for the set of primes in $[y, z)$, and say that n is *z-rough* if it has no prime factor below z. For $\mathcal{C} = \mathcal{A}$ or $\mathcal{B}$ and d whose prime factors are all at least z, $S(\mathcal{C}_d, z)$ is the number of z-rough multiples of d in $\mathcal{C}$. Throughout, $x = q^L$, and the integers of $(x, 2x]$ are those of $(N_1, N_2]$ with $N_1 = \lfloor x \rfloor$ and $N_2 = \lfloor 2x \rfloor$. The tuples $(r_1, \dots, r_j)$ are ordered, and $r = r_1 \cdots r_j$.

The constants. They are chosen before q and depend only on L, u_1, u_2, j, J_0, Λ_1, a_0, b_0 and ε_H. Let $c = (L - 1 - b_0)/4 > 0$. Let $C_0 \ge 1$ be an absolute constant with $\sum_{p \in \mathcal{P}[y, y^2)} 1/p \le C_0$ for every $y \ge 1$; it exists by Mertens' theorem (the library takes it from the prime number theorem). Put $K_1 = (\log((L+1)/a_0) + 1)^j$ and

$$\zeta_k = e^{2C_0} \cdot \frac{2C_0^k}{k!} \cdot \exp\Big(e^{2C_0} \frac{2C_0^{k+1}}{(k+1)!} \Big).$$

Since $(2k+4)^2\zeta_k \to 0$, there is an even $k \geq 2$ with $2C_0^2 \leq (k+1)(k+2)$, $c < (2k+4)a_0$ and $(2k+4)^2\zeta_k \leq \varepsilon_H c^2/(4e(K_1+1)a_0)$. Put $\varepsilon_w = c/(2k+4)$, so that $0 < \varepsilon_w < a_0$ and $(2k+7)\varepsilon_w \leq 2c$, and $S_0 = \lceil L/\varepsilon_w \rceil$. At q, put $w = \lceil q^{\varepsilon_w} \rceil$, $z_0 = \lceil q^{a_0} \rceil$ and $X' = \lfloor q^{b_0} \rfloor$. Then $w \leq z_0$, and by hypothesis every tuple in the sum has $r_i \geq z_0$ and $r \leq X'$.

The tree identity. Let r have all its prime factors at least z_0, and let $r \leq X'$. For a set P of primes write $d_P = r\prod_{p\in P} p$, and let $\rho_z(n) = 1$ if n is z-rough and $\rho_z(n) = 0$ otherwise. Then for every n,

$$\mathbf{1}[r \mid n]\,\rho_{z_0}(n) = \sum_{P\subseteq \mathcal{P}[w,z_0)} (-1)^{|P|}\,\mathbf{1}[d_P \leq X']\,\mathbf{1}[d_P \mid n]\,\rho_w(n)$$
$$+ \sum_{p\in\mathcal{P}[w,z_0)} \sum_{P\subseteq\mathcal{P}(p,z_0)} (-1)^{|P|+1}\,\mathbf{1}[d_P \leq X' < p\,d_P]\,\mathbf{1}[p\,d_P \mid n]\,\rho_p(n), \quad \text{(S3.1)}$$

where $\mathcal{P}(p,z_0)$ is the set of primes in (p,z_0). This is what iterating Buchstab's identity $S(\mathcal{C}_m,z) = S(\mathcal{C}_m,w) - \sum_{w\leq p<z} S(\mathcal{C}_{mp},p)$ from the root r at z_0 gives, when a child mp is stopped as soon as $mp > X'$: the first sum collects the *leaves* $S(\mathcal{C}_d,w)$, $d = d_P \leq X'$, and the second the *stops* $S(\mathcal{C}_W,p)$, $W = p\,d_P$. It is proved directly, for each n. If $r \nmid n$ or n is not w-rough, every term vanishes. Otherwise let Q be the set of the primes of $[w,z_0)$ that divide n; the left side is $\mathbf{1}[Q = \emptyset]$. Since the primes of r are at least z_0, a leaf P counts n exactly when $P \subseteq Q$, and a stop (p,P) exactly when $p = \min Q$ and $P \subseteq Q\setminus\{p\}$. If $Q = \emptyset$, only the leaf $P = \emptyset$ counts n, with $\mathbf{1}[r \leq X'] = 1$. If $Q \neq \emptyset$, let $p_0 = \min Q$; grouping $P \subseteq Q\setminus\{p_0\}$ with $P\cup\{p_0\}$, the right side is

$$\sum_{P\subseteq Q\setminus\{p_0\}} (-1)^{|P|}\big(\mathbf{1}[d_P \leq X'] - \mathbf{1}[p_0 d_P \leq X'] - \mathbf{1}[d_P \leq X' < p_0 d_P]\big) = 0.$$

Multiplying (S3.1) by $\mathbf{1}[n \equiv a] - \mathbf{1}[(n,q) = 1]/\varphi(q)$ and summing over n and over the tuples, the sum of the proposition is the sum over the leaves of $(-1)^{|P|}(S(\mathcal{A}_d,w) - S(\mathcal{B}_d,w)/\varphi(q))$ plus the sum over the stops of $(-1)^{|P|+1}(S(\mathcal{A}_W,p) - S(\mathcal{B}_W,p)/\varphi(q))$.

The stops. A stop has $X' < W \leq pX' < z_0X'$, and W is p-rough, since the primes of r are at least $z_0 > p$ and those of P exceed p. So the p-rough multiples of W in $(N_1,N_2]$ are the $n = Wm$ whose cofactor m is p-rough, and every such m lies in $(M_1,M_2]$ with $M_1 = \lfloor N_1/(z_0X')\rfloor$ and $M_2 = \lfloor N_2/(X'+1)\rfloor$. For large q, $q^{u_1L} \leq M_1$, since $5q^{u_1L+a_0+b_0} \leq q^L$ by $a_0+b_0 < (1-u_1)L$, and $M_2 \leq q^{u_2L}$, since $2q^{L-b_0} \leq q^{u_2L}$ by $(1-u_2)L < b_0$. So the cofactor m lies in the window's range, and W in its reflection. A stop has $|P| \leq S_0$, since $\prod P \geq w^{|P|} \geq q^{\varepsilon_w|P|}$ and $\prod P \leq X' \leq q^L$.

For each $s \le S_0$, the stops with $|P| = s$ form one sum of the kind that Lemma 4.5 bounds, with $m \in (M_1, M_2]$ and $a_m = 1$, whose primes are $r_1, \dots, r_j$, the elements $y_1 > \cdots > y_s$ of P and p, and whose comparisons are the given ones on the r_i and

$$y_1 > \cdots > y_s > p, \quad y_1 < z_0, \quad p \ge w, \quad r y_1 \cdots y_s \le X' < r y_1 \cdots y_s\, p,$$
$$p \le \min\{\ell \text{ prime} : \ell \mid m\}, \qquad N_1 < m r y_1 \cdots y_s p \le N_2.$$

For $s = 0$ the condition $y_1 < z_0$ reads $p < z_0$. Each condition is of the form $\lambda(m) + \sum_i e_i \log x_i > 0$ or ≥ 0, with coefficients in $\{-1, 0, 1\}$ and a pivot. There are at most $J_0 + s + 7$ of them, and $\lambda(m) \le (\Lambda_1 + L + 1)\log q$ for $m \le M_2$. Wherever they all hold, every prime is at least $w \ge 2q^{\varepsilon_w/2}$ and $m r y_1 \cdots y_s p \le 2x$. So Lemma 4.5, with $K_0 = j + S_0 + 2$, $J_0 + S_0 + 7$ comparisons, $e_1 = \varepsilon_w/2$, $E_0 = 1$ and $\Lambda_1 + L + 1$ in place of Λ_1, bounds each of these $S_0 + 1$ sums by $q^{L-\delta_T}/\varphi(q)$, with $\delta_T > 0$ depending only on the constants above. For large q the stops therefore total at most $q^{L-\delta_1}/\varphi(q)$, with $\delta_1 = \delta_T/2$.

A leaf. Let d have all its prime factors at least w. We show

$$\Big| S(\mathcal{A}_d, w) - \frac{S(\mathcal{B}_d, w)}{\varphi(q)} \Big| \le \frac{2\zeta_k (N_2 - N_1)}{d\varphi(q)\log w} + 24\, w^{2k+7}. \tag{S3.2}$$

If $(d, q) > 1$ both counts vanish, so let $(d, q) = 1$. Sieve by the primes below w, in the ranges $T_i = \mathcal{P}[w^{2^{-i-1}}, w^{2^{-i}})$ for $0 \le i < J_w = 2\lfloor \log_2 w \rfloor + 2$, which cover them, at the even levels $k_i = k + 2i$, which grow with i so that the errors $C_0^{k_i}/k_i!$ below have a sum that does not depend on J_w. Brun's sieve [44], in the following form, is used, for disjoint sets T_i of primes and even levels $k_i \ge 2$. Let $\mathcal{C}$ be a finite set of integers, $Y \ge 0$, and g a function on the primes of $\bigcup T_i$ with $0 \le g \le 1$. Suppose that for every choice of $s_i \subseteq T_i$, with $\Pi = \prod_i \prod_{p \in s_i} p$, the count $\#\{n \in \mathcal{C} : \Pi \mid n\}$ is $Y \prod_i \prod_{p \in s_i} g(p)$ within R. Then $\#\{n \in \mathcal{C} : (n, \prod_i \prod_{T_i} p) = 1\}$ is $Y \prod_i \prod_{p \in T_i} (1 - g(p))$ within

$$Y \sum_i e_{k_i}(T_i) \prod_{l \ne i} F_{k_l}(T_l) + (J_w + 1)\, R \prod_i \sum_{m \le k_i} \binom{|T_i|}{m},$$

where $e_m(T) = \sum_{s \subseteq T,\ |s| = m} \prod_{p \in s} g(p)$ and $F_k(T) = \sum_{m \le k} (-1)^m e_m(T)$. Bonferroni's inequalities say that $F_{k-1}(T) \le \prod_{p \in T}(1 - g(p)) \le F_k(T)$ for every even k, and the same for any weights in $[0, 1]$ in place of g. Take them for each $n \in \mathcal{C}$ with the weights $\mathbf{1}[p \mid n]$, and use the elementary inequality $\prod_i \mathrm{hi}_i - \sum_i (\mathrm{hi}_i - \mathrm{lo}_i) \prod_{l \ne i} \mathrm{hi}_l \le \prod_i \varrho_i$ for $\varrho_i \ge 0$ and $\mathrm{lo}_i \le \varrho_i \le \mathrm{hi}_i$ (the lo_i, odd truncations, may be negative). Summed over n, and expanded by the

hypothesis, they place the count, up to (J_w+1) times $R\prod_i \sum_{m\le k_i}\binom{|T_i|}{m}$, in the interval from $Y\prod_i F_{k_i} - Y\sum_i e_{k_i}\prod_{l\ne i}F_{k_l}$ to $Y\prod_i F_{k_i}$. The model $Y\prod_i V_i$, with $V_i=\prod_{p\in T_i}(1-g(p))$, lies in the same interval: Bonferroni's inequalities for g give $F_{k_i}-e_{k_i}=F_{k_i-1}\le V_i\le F_{k_i}$, hence $0\le\prod_i F_{k_i}-\prod_i V_i\le\sum_i e_{k_i}\prod_{l\ne i}F_{k_l}$.

Take $g(p)=1/p$ for $p\nmid q$ and $g(p)=0$ for $p\mid q$. For $\mathcal{A}_d$, take $\mathcal{C}=\{n\in(N_1,N_2]: n\equiv a,\ d\mid n\}$, $Y=(N_2-N_1)/(dq)$ and $R=1$: if no prime of Π divides q, Lemma 4.1 with $M=d\Pi$ gives the count within 1, and otherwise the count and the model both vanish. For $\mathcal{B}_d$, take $\mathcal{C}=\{n\in(N_1,N_2]: (n,q)=1,\ d\mid n\}$, $Y\varphi(q)$ and $R=\varphi(q)$, and divide by $\varphi(q)$. Both main terms are then YV with $V=\prod_{p<w,\ p\nmid q}(1-1/p)$, so that

$$\Big|S(\mathcal{A}_d,w)-\frac{S(\mathcal{B}_d,w)}{\varphi(q)}\Big|\le 2(YG+\mathrm{rem}),$$

$$G=\sum_i e_{k_i}(T_i)\prod_{l\ne i}F_{k_l}(T_l),\qquad \mathrm{rem}=(J_w+1)\prod_i\sum_{m\le k_i}\binom{|T_i|}{m}.$$

Each T_i lies in some $\mathcal{P}[y,y^2)$, so $\sum_{p\in T_i}g(p)\le C_0$ and, since $1-t\ge e^{-2t}$ for $0\le t\le 1/2$, $V_i\ge e^{-2C_0}$. With $e_m\le C_0^m/m!$, $F_{k_l}\le V_l+e_{k_l+1}\le V_l(1+e^{2C_0}C_0^{k_l+1}/(k_l+1)!)$ and $\prod_{l\ne i}V_l\le e^{2C_0}V$, this gives $G\le Ve^{2C_0}\sum_i\frac{C_0^{k_i}}{k_i!}\prod_l\big(1+e^{2C_0}\frac{C_0^{k_l+1}}{(k_l+1)!}\big)$. Since $2C_0^2\le(k+1)(k+2)$, the terms of $\sum_i C_0^{k+2i}/(k+2i)!$ decrease at least geometrically with ratio $1/2$, so the sum is at most $2C_0^k/k!$; likewise, since $2C_0^2\le(k+2)(k+3)$, $\sum_l C_0^{k_l+1}/(k_l+1)!\le 2C_0^{k+1}/(k+1)!$; and with $\prod(1+t_l)\le\exp\sum t_l$ we get $G\le\zeta_k V$. So the difference is at most $2\zeta_k YV+2\,\mathrm{rem}$. Next, $V\le q/(\varphi(q)\log w)$. Indeed $V\prod_{p<w,\ p\mid q}(1-1/p)=\prod_{p<w}(1-1/p)$, the second product on the left is at least $\prod_{p\mid q}(1-1/p)=\varphi(q)/q$, and $\prod_{p<w}(1-1/p)\le 1/\log w$, since $\prod_{p<w}(1-1/p)^{-1}$ is the sum of $1/m$ over the integers $m\ge 1$ whose prime factors are all below w, which is at least $\sum_{m<w}1/m\ge\log w$. So $YV\le(N_2-N_1)/(d\varphi(q)\log w)$. Finally $\sum_{m\le k_i}\binom{|T_i|}{m}\le 2\max(1,|T_i|)^{k_i}$ with $|T_i|\le w^{2^{-i}}$, $\sum_i 2^{-i}(k+2i)\le 2k+4$, $J_w+1\le 3w$ and $2^{J_w}\le 4w^2$ give $\mathrm{rem}\le 12w^{2k+7}$.

The leaves together. In (S3.2), $N_2-N_1\le x+1\le 2x$ and $\log w\ge\varepsilon_w\log q$. With $1\le X'/d$ for the remainder, the leaves total at most

$$\Big(\frac{4\zeta_k x}{\varphi(q)\varepsilon_w\log q}+24\,w^{2k+7}X'\Big)\sum_{(r_1,\dots,r_j)}\ \sum_{P\subseteq\mathcal{P}[w,z_0)}\frac{1}{r\prod P},$$

and the double sum is $\sum 1/r\cdot\prod_{w\le p<z_0}(1+1/p)$. Each r_i is a prime in

$[q^{a_0}, 2q^L]$, so for large q, by Mertens' theorem (in the library, the prime number theorem), $\sum 1/r \le K_1$ and $\prod_{w\le p<z_0}(1+1/p) \le ea_0/\varepsilon_w$. The main part is then at most

$$\frac{4eK_1a_0\zeta_k}{\varepsilon_w^2}\cdot\frac{x}{\varphi(q)\log q} = \frac{4eK_1a_0(2k+4)^2\zeta_k}{c^2}\cdot\frac{x}{\varphi(q)\log q} \le \frac{\varepsilon_H\, x}{\varphi(q)\log q},$$

by the choice of k. Since $w \le 2q^{\varepsilon_w}$ and $(2k+7)\varepsilon_w \le 2c$, the remainder part is at most $24\cdot 2^{2k+7}K_1ea_0\varepsilon_w^{-1}q^{b_0+2c} \le q^{b_0+3c} = q^{L-1-c} \le q^{L-c}/\varphi(q)$ for large q, using $\varphi(q)\le q$.

Conclusion. The sum of the proposition is at most $\varepsilon_H x/(\varphi(q)\log q) + (q^{L-c}+q^{L-\delta_1})/\varphi(q)$, which for large q is at most $\varepsilon_H x/(\varphi(q)\log q)+q^{L-\delta}/\varphi(q)$ with $\delta = \min(c,\delta_1)/2$. □

The spline weights of Section 4

For integers $A < B$, $\ell \ge 1$ and $r \ge 1$ let $w(k)$ be ℓ^{-r} times the number of $(i, j_1, \dots, j_r)$ with $A < i \le B$, $0 \le j_s < \ell$ and $i + j_1 + \cdots + j_r = k$. It is the average over the ℓ^r tuples j of the indicators of the intervals $(A + \sum j_s, B + \sum j_s]$, so $0 \le w \le 1$, $w = 1$ on $(A + r(\ell-1), B]$, $w = 0$ off $(A, B + r(\ell-1)]$, and $\sum_k w(k) = B - A$. Its transform factors: $\hat w(h) = \ell^{-r}\big(\sum_{A<i\le B} e_q(hi)\big)\big(\sum_{j<\ell} e_q(hj)\big)^r$. For h the residue of an integer t with $0 < |t| \le q/2$, a geometric sum of length Y at $e_q(t)$ is at most $q/(2|t|)$ in size, so $|\hat w(h)| \le (B-A)(q/(2\ell|t|))^r$, which is at most $(B-A)(q/(2\ell(H+1)))^r$ when $|t| > H$. If each residue class mod q holds at most M elements of the support, Parseval's identity and $|w_k w_{k'}| \le \frac12(|w_k|^2 + |w_{k'}|^2)$ over the pairs in a class give $\sum_h |\hat w(h)|^2 \le qM\sum_k |w(k)|^2$; for the support, an interval of length $n = B - A + r(\ell-1)$, $M = \lfloor n/q\rfloor + 1$. Finally $|E_w| \le d(q)$: for the indicator of an interval $(C, D]$, Möbius inversion over the divisors of q writes E_w as $\sum_{d|q}\mu(d)\big((\lfloor D/d\rfloor - \lfloor C/d\rfloor) - (D-C)/d\big)$, and each bracket is less than 1 in size; the spline averages such intervals.

Proof of Lemma 4.3

Proof. Write $R(m)$ for the indicator of the integers with no prime factor below w, $\mathrm{rng}(d,m)$ for that of $N_1 < dm \le N_2$, and $c(\mathbf{y}_1, \mathbf{y}_2)$ for the indicator of the comparisons. For d with no prime factor below w, $S(\mathcal{A}_d, w) - S(\mathcal{B}_d, w)/\varphi(q) = \sum_m \mathrm{rng}(d,m)R(m)\,\vartheta(dm)$.

Brun's weights. Sift by the primes below w in the ranges T_i of the proof of Proposition 4.7, at the even levels $k_i = k + 2i$. Bonferroni's inequalities give weights λ^+ above and λ^- below R: $\sum_\tau \lambda^-_\tau \mathbf{1}[f_\tau \mid m] \le R(m) \le \sum_\sigma \lambda^+_\sigma \mathbf{1}[f_\sigma \mid$

$m]$, with moduli $f \le w^{2k+4}$, $\sum|\lambda| \le 2\,\mathrm{rem}$ and $\sum|\lambda|/f \le (3+4\log q)^C$ for a constant C, and the gap Γ between the two weights' main terms, $\sum_\sigma \lambda_\sigma^+ \mathbf{1}[(f_\sigma,q)=1]/f_\sigma - \sum_\tau \lambda_\tau^- \mathbf{1}[(f_\tau,q)=1]/f_\tau$, is at most $\zeta_k q/(\varphi(q)\log w)$ in size, as the gap G of that proof.

The sandwich. Split each λ into its positive and negative parts. With splines $w_- \le \mathrm{rng}(df_\sigma,\cdot) \le w_+$, chosen below for each block, the count of $\mathcal{A}$ is at most $\sum_\sigma(\lambda_\sigma^+)_+ \sum_k w_+(k)\mathbf{1}[df_\sigma k \equiv a] - \sum_\sigma(\lambda_\sigma^+)_- \sum_k w_-(k)\mathbf{1}[df_\sigma k \equiv a]$, and at least the same expression with the weights λ^- and the splines exchanged; and the same holds for $\mathcal{B}/\varphi(q)$. The difference between the $\mathcal{A}$- and the $\mathcal{B}/\varphi(q)$-counts is thus bounded by two kinds of term: the difference of the main terms, which carries the gap Γ and the difference of the splines' masses, and bilinear sums $\sum\lambda\sum_k w(k)\,\vartheta(dfk)$.

Blocks. For $\rho = 1+q^{-s}$ group the pairs (d_1f, d_2), $d_1 = \prod \mathbf{y}_1$ and $d_2 = \prod \mathbf{y}_2$, by the blocks $\lfloor \log(d_1f)/\log\rho \rfloor$ and $\lfloor \log d_2/\log\rho \rfloor$; a block pair (β_1,β_2) puts d_1fd_2 in $[\rho^{\beta_1+\beta_2}, \rho^{\beta_1+\beta_2+2})$. At the scale $D' = \rho^{\beta_1+\beta_2}$ one majorant w_+, equal to 1 on $(\lfloor N_1/(D'\rho^2)\rfloor, \lfloor N_2/D'\rfloor]$, and one minorant w_-, supported in $(\lfloor N_1/D'\rfloor, \lfloor N_2/(D'\rho^2)\rfloor]$, both splines of step $\ell = \lfloor \eta N_1/(rD'\rho^2)\rfloor + 1$ with $\eta = q^{-s}$, sandwich $\mathrm{rng}(d,\cdot)$ for every d of the block pair, and their masses differ by at most $(N_1+N_2)(1-\rho^{-2})/D' + 2\eta N_1/(D'\rho^2) + 2$.

A block pair. On a block pair Lemma 4.2 applies, with $\mathcal{D}_1$ the values d_1f, whose coefficients aggregate the fibres $\{(\mathbf{y}_1,\sigma) : \prod\mathbf{y}_1 \cdot f_\sigma = d_1f\}$ (at most $(J_w+1)d(d_1f)$ elements, by the uniqueness of the factorisation), $\mathcal{D}_2$ the values d_2 (a decreasing tuple of primes is determined by its product), $P_\iota = \rho^{\beta_\iota+1}$, $D = \lfloor P_2\rfloor$, $H = \lfloor (q-1)/(2D)\rfloor$, so that $2HD < q$ and $q/(2\ell(H+1)) \le D/\ell$, and $m = \lfloor P_1/q\rfloor + 1$, since the values $d_1f \le P_1$ with a given inverse lie in one residue class. The comparisons are replaced by smooth cut-offs of their forms, as in the proof of Lemma 4.4 (§S2): the smooth part is separated by Fourier inversion into bilinear forms of the lemma's kind, at a cost of the cut-offs' Fourier masses, and the part near the boundaries is treated as in §S2: with $|\vartheta(n)| \le \vartheta(n) + 2\cdot\mathbf{1}[(n,q)=1]/\varphi(q)$, the part with ϑ is again a bilinear form of the lemma's kind, with the narrow cut-offs, and the part with $2/\varphi(q)$ is counted along the pivots. A block pair is so bounded by its Fourier masses and fibre sizes times

$$\sqrt{P_1P_2(P_1/q+1)F(n/q+1)n} + (B-A)(P_2/\ell)^r P_1P_2 + P_1P_2\,d(q)/\varphi(q),$$

with n the length of the spline, plus the counted parts near the boundaries.

The bound in powers of q. Take $s = \min\big((\kappa_1-\kappa_0)/(2(J_0+6)),\ \kappa_2/2,\ \varepsilon_w,\ 1/(4(J_0+6))\big)$, $r = \lceil 4/\kappa_2\rceil$ and $F = \lfloor C_d q^s\rfloor$, a bound for the number of divisors of the integers below q. On a block pair that

holds a leaf, with $X_1 = q^{L-1-\kappa_1}$, $X_2 = q^{L-\kappa_2}$ and $X_3 = q^{L-1-\kappa_0}$ in the lemma's conditions, the three terms are at most $\sqrt{816C_d}\,q^{L-1-(\kappa_1-\kappa_0-s)/2}$, $16(16r)^r q^{L-2}$ and $4C_d q^{L-1/2+s}/\varphi(q)$; a block pair that holds no leaf contributes nothing. There are at most $(3Lq^s\log q)^2$ block pairs, the Fourier masses of at most J_0 cut-offs are at most $(J_0+1)(Cq^s)^{J_0}$, and the fibres at most q^{c_1} with $c_1 < s/4$; so the bilinear sums total at most $C(\log q)^2 q^{L-3s/4}/\varphi(q)$, by the choice of s.

The main terms. The difference of the main terms is $(N_2-N_1)/q\cdot\Gamma\cdot\sum c/d$, plus the transition from the splines' masses, at most $(V\Theta/q)\sum c/d + (4W/q + 2Wd(q)/\varphi(q))\sum c$ with $\Theta \le 11q^{L-s}$, $V = \sum|\lambda|/f$ and $W = \sum|\lambda|$. Since $(d_1d_2)^2 \le X_1X_2$ on the leaves, $\sum c \le q^{L-1/2}B^{K_1+K_2}$, with B bounding the sums of $1/p$ over the primes' ranges; and the counted parts near the boundaries are at most $(12\theta + 1/w)B^{K_1+K_2-1}$ times $2(1+\eta)\rho^2 N_2 V/\varphi(q)$, by the pivots, since every prime is at least w, with $\theta = q^{-s}$. All these are at most $C(\log q)^2 q^{L-3s/4}/\varphi(q)$, which is below $q^{L-s/4}/\varphi(q)$ for large q. With $|\Gamma| \le \zeta_k q/(\varphi(q)\log w)$ this is the lemma, with $\delta = s/4$. □

Proof of Proposition 4.8

Proof. The notation is that of the proof of Proposition 4.7.

The constants. Let $c' = \min\bigl(1/16,\ (L-2)/16,\ (L-1-D_1)/16,\ (L-D_1-2D_2)/16\bigr) > 0$. Take an even k as in that proof, with c' in place of c, and put $\varepsilon_w = c'/(2k+4)$, so that $(2k+4)\varepsilon_w = c' \le 1/16$, and $s_{\max} = \lceil (L+2)/\varepsilon_w\rceil$. At q, put $w = \lceil q^{\varepsilon_w}\rceil$ and $z_0 = \lceil q^{a_0}\rceil$, and let $(P_1,P_2] = (\lceil q^{u_1L}\rceil, \lfloor q^{u_2L}\rfloor]$ be the window's range and $(W_0,W_1] = (\lceil N_2/q^{u_2L}\rceil, \lfloor N_1/\lceil q^{u_1L}\rceil\rfloor]$ its reflection, where Lemma 4.6(c) applies.

The patterns. Split the tuples by the position of $r_{\mathcal{S}_0} = \prod_{i\in\mathcal{S}_0} r_i$: $r_{\mathcal{S}_0} \le P_1$, $r_{\mathcal{S}_0}$ in the window's range, $P_2 < r_{\mathcal{S}_0} \le W_0$, $r_{\mathcal{S}_0}$ in the reflection, $r_{\mathcal{S}_0} > W_1$. The last is empty for large q, since $r_{\mathcal{S}_0} \le q^{\sigma^+}$ and $\sigma^+ < (1-u_1)L$. Each of the others is a family of the same kind, with two more comparisons.

$r_{\mathcal{S}_0}$ in the window or its reflection. By Buchstab's identity, $S(\mathcal{A}_r, z_0) = S(\mathcal{A}_r, r_j) + \sum_{z_0\le p<r_j} S(\mathcal{A}_{rp}, p)$, and the same for $\mathcal{B}$; both sums are of the kind of Lemma 4.6, (b) with $\mathcal{I} = \mathcal{S}_0$ or (c) with $\mathcal{J} = \mathcal{S}_0$, the second with the chain extended by p. So these patterns contribute at most $2(q^{L-\delta_1} + q^{L-\delta_2})/\varphi(q)$.

$r_{\mathcal{S}_0}$ below the window, or between it and its reflection. Let $E = P_1 + 1$ in the first case and $E = W_0 + 1$ in the second. Iterate Buchstab's identity from z_0 down to w, stopping a branch as soon as $r_{\mathcal{S}_0}$ times its primes reaches E. As in (S3.1), with the condition $r_{\mathcal{S}_0}\prod P < E$ in place of $d_P \le X'$, this writes $S(\mathcal{A}_r, z_0)$ as the sum over the *leaves*, the sets P of primes of $[w, z_0)$ with $r_{\mathcal{S}_0}\prod P < E$, of $(-1)^{|P|}S(\mathcal{A}_{r\prod P}, w)$, plus the sum over the *stops*,

the pairs (p, P) with P a leaf of primes above p and $r_{\mathcal{S}_0} \prod P \cdot p \geq E$, of $(-1)^{|P|+1} S(\mathcal{A}_{r \prod P \cdot p}, p)$; the identity is proved for each n in the same way, since the products $r_{\mathcal{S}_0} \prod_{p \in P,\, p \geq y} p$ of the upper parts of a set grow by factors below z_0 as y decreases, so that a set with $r_{\mathcal{S}_0} \prod P \geq E$ has exactly one first upper part that stops. A stop has $E \leq r_{\mathcal{S}_0} \prod P \cdot p < E z_0$, in the window's range, or in its reflection, for large q, since $a_0 < (u_2 - u_1)L$. So for each size of P, at most $s_{\max}$ of them, the stops form one sum that Lemma 4.6(b), or (c), bounds, with $\mathcal{S}_0$ and the primes of P and p as $\mathcal{I}$, or $\mathcal{J}$.

The leaves. A leaf has $\prod P < E / r_{\mathcal{S}_0} \leq 4q^t$ for large q, by the hypotheses on t, since $r_{\mathcal{S}_0} \geq q^{\sigma^-}$, and $r_{\mathcal{S}_0} > P_2$ in the second case; and $d = r \prod P$. Those with $d \leq q^{L-1-4c'}$ are counted by (S3.2), and total at most $(2\zeta_k(N_2 - N_1)/(\varphi(q) \log w) + 24 w^{2k+7} q^{L-1-4c'}) \sum 1/d$. The others are split into d_1, the product of the r_i with $i \in \mathcal{S}_1$ and, if $\iota = 1$, of the primes of P, and d_2, the rest. Then $d_1 \leq 4q^{D_1}$ and $d_2 \leq 4q^{D_2}$, so $d_1 w^{2k+4} \leq q^{L-1-8c'}$ and $d_1 w^{2k+4} d_2^2 \leq q^{L-8c'}$ for large q, by the choice of c'; the primes of P are below $z_0 \leq r_j$, so the merged tuples are strictly decreasing; and the comparisons with no nonzero coefficient, constant on the family, can be dropped. So Lemma 4.3, with $\kappa_1 = \kappa_2 = 8c'$ and $\kappa_0 = 4c'$, bounds the large leaves, for each pair (K_1, K_2) of tuple lengths, by $\zeta_k(N_2 - N_1)/(\varphi(q) \log w) \sum 1/d + q^{L-\delta_3}/\varphi(q)$.

Conclusion. As in the proof of Proposition 4.7, $\sum 1/d \leq K_1 e a_0 / \varepsilon_w$ for large q, so the main parts of the leaves total at most $\varepsilon_H x/(\varphi(q) \log q)$ by the choice of k, and the remainders are at most $q^{L-c'}/\varphi(q)$. The sum of the proposition is therefore at most $\varepsilon_H x/(\varphi(q) \log q) + q^{L-\delta}/\varphi(q)$ for large q, with $\delta = \min(c', \delta_1, \delta_2, \delta_3)/2$, δ_1 and δ_2 those of Lemma 4.6, (b) and (c), for these families, and δ_3 that of Lemma 4.3. □

S4 Proof of Lemma 5.1

Proof. Write $c_* = 0.565073$ and $\bar{\omega} = 0.5671433$, $u = \log 2y / \log z$, and say that an integer is z-rough if it has no prime factor below z. Four facts are used.

(a) Prime sums. By the prime number theorem with an error $o(t/\log t)$ (in the library, the bound $\psi(t) - t = O(t^{11/12+\varepsilon})$, which it derives from the case of ζ of Input 1; the classical theorem would do), for every $\varepsilon_3 > 0$, $|\theta(t) - t| \leq \varepsilon_3 t / \log t$ for large t. With partial summation this gives, for large $a \leq b$,

$$\sum_{a < p \leq b} \frac{1}{p \log p} \leq \frac{1}{\log a} - \frac{1}{\log b} + \frac{4\varepsilon_3}{\log^2 a},$$

and, if $2\log b \le \mathcal{L}$,

$$\sum_{a<p\le b}\frac{1}{p(\mathcal{L}-\log p)} \le \frac{1+\varepsilon_3/\log a}{\mathcal{L}}\log\frac{\log b\,(\mathcal{L}-\log a)}{\log a\,(\mathcal{L}-\log b)} + \frac{6\varepsilon_3}{\mathcal{L}\log^2 a};$$

and, for large y, there are at most $(1+\varepsilon_3)y/\log 2y$ primes in $(y,2y]$.

(b) Buchstab's inequality. For $y \ge 2$,

$$\Phi(y,z) \le \#\{p : y < p \le 2y\} + \sum_{z\le p\le\sqrt{2y}}\Phi(y/p,p),$$

since a z-rough integer $m \in (y,2y]$ that is not prime has its least prime factor p in $[z,\sqrt{2y}]$, and m/p is a p-rough integer in $(y/p,2y/p]$; the map $m \mapsto (p,m/p)$ is injective. Also $\Phi(y,z)=0$ if $y\ge 1$ and $2y<z$, since every integer of $(y,2y]$ is then at least 2 and below z.

(c) Buchstab's function on $[2,3]$. For $u>1$, $1+\log(u-1)\le\bar\omega u$: the inequality $\log t \le t-1$ at $t=\bar\omega(u-1)$ gives $1+\log(u-1)\le\bar\omega(u-1)-\log\bar\omega\le\bar\omega u$, since $-\log\bar\omega\le\bar\omega$, that is $\bar\omega e^{\bar\omega}\ge 1$, which holds because $e^{\bar\omega}\ge\sum_{m\le 12}\bar\omega^m/m! > 1/\bar\omega$. So $\omega^*\le\bar\omega$ on $[2,3]$, and since $c_*<\bar\omega$, $\omega^*\le\bar\omega$ on $[2,\infty)$; and $\omega^*\le c_*$ on $[3,\infty)$, since $\omega^*(3)=(1+\log 2)/3<c_*$.

(d) A delay inequality. Let $M(u)=\log(u-1)$ for $2\le u\le 3$ and $M(u)=\log 2+\bar\omega\min(u-3,1)+c_*\max(u-4,0)$ for $u>3$, which is $\int_1^{u-1}\omega^*(t)\,dt$ with ω^* replaced by $\bar\omega$ on $[2,3]$. Then $1+M(u)\le u\omega^*(u)$ for $u\ge 2$. For $u\le 3$ it is an equality. For $u>3$, $\omega^*(u)=c_*$, and $1+M(u)-c_*u$ is affine on $[3,4]$, with slope $\bar\omega-c_*>0$, and constant on $[4,\infty)$; so it suffices that it is at most 0 at $u=4$, that is $1+\log 2+\bar\omega\le 4c_*$, which holds since $\log 2<0.6931471808$ and $2.2602904808<2.260292=4c_*$.

We prove by induction on the integer $k\ge 0$ the statement C(k): for every $\varepsilon>0$ there is z_1 such that $\Phi(y,z)\le(\omega^*(u)+\varepsilon)y/\log z$ for all $z\ge z_1$ and all $y\ge 1$ with $u\le k$. C(0) holds trivially, since $u>0$, and the lemma is C($\lceil U\rceil$). Assume C(k), let $0<\varepsilon\le 1$ (which suffices, since the bound for one ε implies it for every larger one) and $\varepsilon_3=\varepsilon/100$, let z be large in terms of ε and of the z_1 of C(k) for $\varepsilon/4$, and let $y\ge 1$ with $\mathcal{L}=\log 2y=u\log z\le(k+1)\log z$. If $2y<z$, then $\Phi(y,z)=0$. Otherwise $u\ge 1$, $y\ge z/2$ is large, and (a) and (b) give

$$\Phi(y,z)\le\Big(1+\frac{\varepsilon}{4}\Big)\frac{y}{\mathcal{L}}+\sum_{z\le p\le\sqrt{2y}}\Phi(y/p,p).$$

If $u<2$, then $\sqrt{2y}<z$, the sum is empty, and $\Phi(y,z)\le(1+\varepsilon/4)y/(u\log z)\le(\omega^*(u)+\varepsilon)y/\log z$. Let $u\ge 2$. A prime $p\in[z,\sqrt{2y}]$ has $y/p\ge 1$ and $\log(2y/p)=\mathcal{L}-\log p\le k\log p$, since $\mathcal{L}\le(k+1)\log p$. So C($k$)

gives $\Phi(y/p,p) \le (\omega^*(u_p)+\varepsilon/4)\,y/(p\log p)$ with $u_p = (\mathcal{L}-\log p)/\log p$. The term $p=z$, if z is prime, contributes at most $2y/(z\log z)$, since $\omega^* \le 1$. By (a), the parts $\varepsilon/4$ of the other terms contribute at most $(\varepsilon/4)(1/\log z + 4\varepsilon_3/\log^2 z)\,y$. For the parts $\omega^*(u_p)$: if $p \le (2y)^{1/4}$ then $u_p \ge 3$ and $\omega^*(u_p) \le c_*$; if $(2y)^{1/4} < p \le (2y)^{1/3}$ then $2 \le u_p < 3$ and $\omega^*(u_p) \le \bar\omega$; and if $(2y)^{1/3} < p \le (2y)^{1/2}$ then $1 \le u_p < 2$ and $\omega^*(u_p)/(p\log p) = 1/(p(\mathcal{L}-\log p))$, by the definition of ω^* on $[1,2)$. By (a), applied on $(z,\sqrt{2y}]$ if $u \le 3$, on $(z,(2y)^{1/3}]$ and $((2y)^{1/3},\sqrt{2y}]$ if $3 < u \le 4$, and also with the first range split at $(2y)^{1/4}$ if $u > 4$,

$$\sum_{z<p\le\sqrt{2y}} \frac{\omega^*(u_p)}{p\log p} \le \frac{M(u)+12\varepsilon_3}{\mathcal{L}} + \frac{8\varepsilon_3}{\log z}.$$

Collecting the terms, with $\mathcal{L} = u\log z$,

$$\Phi(y,z) \le \frac{y}{\log z}\Big(\frac{1+M(u)}{u} + \frac{\varepsilon/4+12\varepsilon_3}{u} + \frac{2}{z} + 8\varepsilon_3 + \frac{\varepsilon}{4} + \frac{\varepsilon\varepsilon_3}{\log z}\Big).$$

By (d) the first term is at most $\omega^*(u)$, and since $u \ge 1$ and z is large the others add up to at most 0.96ε. So $\Phi(y,z) \le (\omega^*(u)+\varepsilon)y/\log z$, which proves $\mathrm{C}(k+1)$. □

A fact used in Section 5. The bound Ω of that section, though not the proof above, uses that on $[2,3]$, $\omega^*(u) = (1+\log(u-1))/u$ increases up to 2.76 and decreases from 2.77 on. For $2 \le u \le u' \le 2.76$, $\omega^*(u) \le \omega^*(u')$ is $(u'-u)(1+\log(u-1)) \le u\log((u'-1)/(u-1))$: the right side is at least $u(u'-u)/(u'-1)$, by $\log t \ge 1-1/t$, and the left at most $(u'-u)u/1.76$, by $\log t \le t-1$ at $t = (u-1)/1.76$ and $\log 1.76 \le 1/1.76$. For $2.77 \le u \le u'$ the reverse inequality follows in the same way from $\log((u'-1)/(u-1)) \le (u'-u)/(u-1)$ and $\log(u-1) \ge \log 1.77 \ge 1/1.77 \ge 1/(u-1)$. The two constants come from the exponential's series.

S5 The formalisation in detail

This section completes Section 7 of the paper.

Chen, Gupta and Li's paper

For Chen, Gupta and Li's paper [12] the library formalises its Proposition 3.1, in the form for one point per character of Section 2 of the paper (`LinnikCGLLV.CGLProp31`, proved as `LinnikCGLHB101.prop31`), with

the results it rests on (their Propositions 4.6, 6.1, 8.1 and 11.1 and Theorem 10.1), the steps of Input 2, and the estimate with exponent $7/3+\varepsilon$ on $[5/7, 16/21]$ (`LinnikCGLDensity.cglDensity`); their Theorem 1.2 as they state it and their other results are not formalised. The library's proof of their Theorem 10.1, a large-values bound with characters that they credit to Heath-Brown [14], follows the arguments of Kerr [15] and Jutila [16]; some lemmas of their paper were formalised from an outline of their proofs; and the formalisation found six places where the text of their paper needs a small repair, none affecting its results, which the author has sent to them (`formalisations/chen-gupta-li-density.md`).

The statements

The statement of Input 2 and those of the density estimates count the characters χ mod q, as elements of Mathlib's `DirichletCharacter` $\mathbb{C}$ `q`, for which `LFunction` χ `s = 0` at some s in the rectangle. Mathlib's L-function of a character mod q is the analytic continuation of $\sum_n \chi(n)n^{-s}$, with the character taken mod q and not reduced to a primitive one. So $L(s,\chi)$ is the L-function of the primitive character inducing χ times the Euler factors at the primes dividing q, whose zeros lie on $\operatorname{Re} s = 0$ and do not affect the counts for $\sigma \geq 1/2$. In Lean the excess of Ingham's steps [17] is 4ε for the ε of `LinnikInghamC.charDensity_ingham_c`, whose exponent is $2+4c+4\varepsilon$, with c chosen so that $2+4c = 3/(2-\sigma_+)$, Ingham's exponent at the step's right end σ_+: $\varepsilon = 1/8000$ on the first nineteen steps and $\varepsilon = 1/400000$ on the last. Huxley's estimate is proved in the library in its own form, for the characters mod q (`formalisations/huxley-density.md`). `formalisations/` records, for each result from the literature, the differences between the Lean statement and the published one.

The proof in Lean

The statements of Sections 4 and 5 carry, in Lean, the instances `SieveRoute.Route`, which holds L and the window's ends as the fractions $w_1 = u_1$ and $w_2 = u_2$ of L, and `SieveRoute.RouteOK`, which holds $2 < L \leq 5/2$ and $0 < u_1 < u_2 < 1$, and the hypothesis `SieveRoute.TypeIIAt`, the conclusion of Theorem 3.1 at this L on the window $[u_1 L, u_2 L]$. In the same way a statement of the paper that holds under an input either takes what the input supplies as a hypothesis (`SieveTypeIIHM.typeII_gen_hm`, Proposition 3.5, takes the counts of a density curve, which Proposition 2.2 supplies from Input 2) or uses the

library's proof of the input. The exception in Input 1 is the pole of $L(s, \chi_0)$ at $s = 1$, where Mathlib's function, like every function in Lean, still has a value; the Lean statement leaves that point aside. The curve g^* is `SieveCurve7D.g7d`, and its exponents are written as fractions; the first step's is $3/(2 - 81/140) + 1/2000 = 840199/398000$.

In Section 3, the Halász–Montgomery inequality is used in the form over the characters mod q at one height, `SieveTypeIIHM.sum_bin_hm`, a case of `LinnikPower.hm_duality`, which allows one point for each character (`formalisations/halasz-montgomery.md`); the Lindelöf-type bound behind the kernel bound is `LinnikLindelof.lindelof`. The Lean definition of admissible windows, `SieveCurveGlue.WindowOKHM`, has the form of Definition 3.3. The 134 inequalities of the proof of Proposition 3.6 are among the 1080 corners that `tools/linnik/route_limits.py` checks in exact rationals and the Lean checks by `norm_num`. The Lean (`SieveTypeIIU22000D.window_22000d`) uses neither the convexity nor the monotonicity of that proof. On each of the 66 pieces below 5/6 it checks 16 corners: β at both ends of the piece, $\varpi \in \{0, 4 \cdot 10^{-6}\}$, and the form of E_1 for $\mu \le 1$ at $\mu \in \{\mu_1, 1\}$ and its form for $\mu \ge 1$ at $\mu \in \{1, \mu_2\}$. On the top piece it checks 24: E_2 and E_3 at $\mu \in \{\mu_1, 1\}$ and E_1 at $\mu \in \{1, \mu_2\}$. In all $66 \cdot 16 + 24 = 1080$.

In Sections 4 and 5, `formalisations/harman-sieve.md` gives the sources of Proposition 4.7 and Lemma 5.1 and the differences between the Lean statements and the classical ones, and `formalisations/README.md` lists the Fourier separation of variables behind Lemmas 4.4 to 4.6 among the standard results. The sifting function $S(\mathcal{C}_r, z)$ is, in the Lean, the number of the multiples of r in $\mathcal{C}$ with no prime factor below z (`SieveTree.rough`), and so agrees with the paper's when r has no prime factor below z, as in every use; and the count $\Phi(y, z)$ of rough numbers is `SieveRough.Psi`. The comparisons of Lemma 4.6 are `SieveChain.GCmp`, with coefficients in $\{-1, 0, 1\}$. `SieveProp6.prop6` allows real coefficients in $[-1, 1]$ with a pivot (`SieveCmp.Cmp`), and Proposition 4.7 states its case for comparisons with coefficients in $\{-1, 0, 1\}$, which is the one the decomposition uses (`SieveLeaves.P6_spec`, through `SieveChain.pivoted`). Proposition 4.8 is `SieveP6ib.prop6ib`, used by the decomposition through `SieveLeaves.P6IB_spec` and `SieveQ.P6IB_boxes`, which turn the witnesses of (R2) (`SieveCert.ibOK_sound`) into its hypotheses; the bilinear estimate is `SieveTypeIB.typeIB_bound`, its form with a spline weight `SieveSpline.typeIB_spline`, and Lemma 4.3 is `SieveP6Leaves.leaves_chain`, the case for Brun's weights of the asymptotic form `SieveTypeIBAsym.leaves_ib`. The rule is `SieveCert.ibRule`,

(R1) or the bilinear test with no boundary term (`SieveCert.okIB_cases`), at an exponent that the certificate takes as a parameter (`SieveCert.IBPar`) and `Linnik22000E.ibFits22000E` identifies with L. The library tests the rules of the decomposition in a fixed order: at depth two (E) first, the empty cells of (O) being dropped with the corner cells, at no cost; at depths three and five (O) first; at depths four and six $\alpha_k \geq \alpha_{k-1}$ first, then (E), then the rest of (O); and at depths seven and eight (O) alone.

In Lean (`SieveMain8.linnik_route_lt8`) the grid conditions of Theorem 6.1 are `SieveCert.GridOK`, which holds $2U_2 < D$, $2e_{N-1} < 1 < 2(v+Nh)$, $4h \leq v$, $0 < h$ and $U_1 < U_2$; `SieveCert.GridOKC`: `GridOK` for the coarse grid, $4h + 2h_C \leq v$ and $Nh \leq N_C h_C$; and `SieveDrop8.DeepGrid`: $h_C = 4h$, $h > 0$ and $4h + 4h_C \leq v$. The Lean's grids also carry a multiplier, `SieveCert.Grid.mult`, the side being rounded up to a multiple of the multiplier over D; Theorem 6.1 is the case of the multiplier 1. With $h_C = 4h$ the coarse cells are unions of four cells, $e^C_{N_C-1} = e_{N-4}$ and $N_C h_C = Nh$, so beyond `GridOK` these ask only $4h + 4h_C = 20h \leq v$, which implies $16h \leq v$ and $4h + 2h_C = 12h \leq v$. The condition $2 < L$ of the theorem is a field of `SieveRoute.RouteOK`; it follows from the others, since $LU_2 > D > 2U_2$. The grid conditions of the grid of Proposition 5.4 are checked by `decide` (`SieveCert.gridOK22000E`, `SieveCert.gridOKC22000E` and `Linnik22000E.deepGrid22000E`). The prime number theorem of Sections 4 to 6 is derived from the case of ζ of Input 1, in the form $\psi(t) - t = O(t^{11/12+\varepsilon})$ (`HalfPlanePNT.psi_isBigO_eleven_twelfths`).

The library's proof of Input 1

The library proves Input 1 as `Eis.LFunction_ne_zero_of_re_gt`, along the route of OpenAI's companion preprint [10], on its own toolchain; it imports nothing from OpenAI's release (`formalisations/openai-half-plane.md`). The route has five steps. Power savings for the smoothed Möbius sums of the Dedekind zeta function of $\mathbb{Q}(\sqrt{-3})$ give the half-plane. The target is embedded in a family of sextic twists, over which a mean square bounds it. Poisson summation and the Gauss–Jacobi identities turn the Möbius function times sextic Gauss sums into cubic Gauss sums. Kubota's cubic theta function [32] supplies the transformation, with Patterson's formula for its coefficients [33], Dunn and Radziwiłł's cusp expansions [34], and the quadratic large sieve over $\mathbb{Z}[\omega]$ [35, 36], which bounds the dual mean square. Möbius inversion removes the cube factors and gives the exponent 11/12. The chain gives ζ and $L(s, \chi_{-3})$ first, then the real characters, the characters of order dividing 6, and every other character through the family twisted by a

character modulo $4q$. The classical results that the route uses, among them cubic and sextic reciprocity [37], are theorems of the library. The import cone of the theorem in the library has 228 files and 108696 lines of Lean, all of them in `lib/core`.

Independent checking of the inputs

Independent checking of Input 1 is partial, and little of it is independent of OpenAI. OpenAI's release has its own Lean formalisation of the main preprint's result. Its catalogue `lean/formalization.yaml` [11], read on 25 October 2026, lists that preprint among the papers with a formalised main result, and names the theorem `OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`. Its statement is for every modulus $q \geq 1$, every character mod q and every s with $\operatorname{Re} s > 7/8$, against Mathlib's L-function, with the same exception at $s = 1$ for the trivial character, so it implies Input 1 as stated here. It is OpenAI's own record, not an independent one, and the catalogue gives its scope as partial progress and its review status as unchecked. The author read that release's statements, scanned its Lean and rebuilt it on 7 October 2026, which showed its two main theorems to depend on Lean's standard three axioms only, but did not read its proofs in full. This library's proof of Input 1 is a new formalisation of the companion preprint's route, on a different Lean and Mathlib; it imports nothing from the release and was not compared with it. It is a machine-checked proof of the companion preprint's theorem, but the route is OpenAI's and the formalisation is the author's, written with the AI assistance described in the paper's acknowledgements, so it is not a review of the mathematics that is independent of OpenAI and of the author. The check recorded by the tracker AIProofWatch [50] on 25 October 2026 concerns the specialisation to the Riemann zeta function at $7/8$, not Dirichlet L-functions in general. So the author knows of no review of the mathematics for Dirichlet L-functions in general that is independent of OpenAI and of the author.

For Input 2 the author knows of no independent check either. The library's proof of Chen, Gupta and Li's Proposition 3.1 is the author's formalisation of their text, written in the same way. Lean's kernel checks that proof; it does not check that its statement is theirs.

What the kernel checks

The library's own sources, the directories `lib/core`, `lib/pnt`, `lib/dh`, `lib/zeta23` and `lib/check`, contain no `sorry`, no declared axiom and no `native_decide`, which `scripts/check_hygiene.py` checks. The library vendors 19 files of PrimeNumberTheoremAnd [51] and, in `lib/vendor/zeta23`, 59 files of the `zeta23` folder of Anthropic's `formal-math` [52], each ported to the library's toolchain by a patch that the repository records. That scan does not cover them; but the import cone of Theorem 1.1 contains 15 of the PrimeNumberTheoremAnd files and none of the `zeta23` ones, and not `Wiener.lean`, the vendored file that contains `sorry`. In any case the axiom check of the theorems in the table of Appendix A covers the whole dependency cone of each, and a `sorry` in it would appear there as `sorryAx`.

The certificate of Proposition 5.4 is checked by kernel evaluation, `decide +kernel`, and not by compiled code, split into 1136 parts in 114 files. The 272 rows of the grid form 229 ranges of consecutive rows; 32 of the ranges are parts, and each of the other 197, the heaviest single rows, is cut into pieces of columns, 1104 in all. Each part proves that its sum is at most a bound $\lceil 1.001 \times \text{part} \rceil$, computed exactly, and the bounds sum to $1075756859466551505 < 2^{60}$. The exact total, $\Sigma = 1074682177289261666$, is computed from the same definitions by `tools/linnik/mirror8.py`, and `tools/linnik/gen_cert8.py` cuts it into the parts by the kernel's predicted time. The kernel's check of the 114 files took 6.2 hours of one core in all, and 1.6 hours on four cores, with at most 2.8 GiB for any one file. The definitions the kernel evaluates are arithmetic on Lean's core natural numbers, lists and booleans; their files import from Mathlib only the notation $\mathbb{N}$. Lean's kernel reduces the operations of the natural numbers on literals (addition, multiplication, division, comparison and the like) by the GMP library's arithmetic rather than by their definitions, so this check trusts GMP as well as the kernel.

The artefact

The library, the paper and this supplement are in the repository [49]; this version of the paper and of the supplement describes the library at the commit that contains it. The earlier versions of the paper proved the exponent 278/125, by the decomposition to depth six with kept parts and without the bilinear estimate; that theorem, `Linnik22240R.linnik_22240r`, with its certificate, window and Type II estimate, is in the repository's history,

at commit `8ba053b`. The toolchain is Lean `v4.35.0-rc3` [47], a release candidate, with Mathlib [48] at commit `0f64d30a`, PrimeNumberTheoremAnd at `650d312` and `formal-math` at `fbdc36b`. The repository's README gives the commands: `lake exe cache get` for Mathlib's prebuilt files (which the build then trusts; building Mathlib from source, or replaying the environment with `lean4checker`, removes that trust), `scripts/build.sh` for the build, which limits how many of the memory-hungry kernel checks of the certificate run at once, and `python3 scripts/check_axioms.py` for the axioms; the repository's continuous integration runs them on every change to the library.

S6 Computations around the exponent

This section gives the computations behind the room that the window of Section 3.6 leaves, behind the remark there on the Halász–Montgomery branch, and behind the last two paragraphs of Section 8 of the paper. They are computations with the definitions of the window and of the certificate, by the scripts of `tools/linnik/`, and not theorems of the library; nothing in the paper or the library depends on them.

The room in the window

The window of Proposition 3.6 has little room. All the corners are exact rational inequalities, so there is no numerical tolerance. With u_1 and u_2 fixed, they hold exactly for $L \geq 2.1999832\ldots$, so at 219999/100000 but not at 219998/100000 (`route_limits.py corners g7d 219999/100000 43457 46959 100000`); raising the first step's exponent by more than $4.4404 \cdot 10^{-5}$ moves the lower end out. The Type II estimate on this window does not depend on L otherwise, and Theorem 6.1 also needs $2 < L \leq 5/2$ and $LU_2 > D$, which still hold. The certificate depends on L only through the bilinear test's condition $a_1 < 1 - 1/L$, which the block test also applies and which is stricter for smaller L; on the grid of Proposition 5.4 its total at $L = 137499/62500 = 2.199984$ is the same integer as at $11/5$ (`mirror8.py 43457 46959 272 137499/62500`). So with this window the route also holds at 2.199984, about $1.6 \cdot 10^{-5}$ below $11/5$; a window chosen afresh for a smaller L goes further (below). The corners' slack grows along the narrow steps, to $8.6 \cdot 10^{-4}$ on the fourteenth, and from there steps of $1/280$ leave the window as it is: the least slack of a corner on them is $6.8 \cdot 10^{-4}$, on the first of them, and on the last step, $[163/210, 131/168]$, which contains $7/9$, it is $3.95 \cdot 10^{-3}$. Steps of $1/140$ in place of those of $1/280$, from the end of the fourteenth

step, $5/7 + 1/120$, would leave the window as it is too, with a least slack of $3.8 \cdot 10^{-4}$ on them; at $278/125$, with the window of the earlier version, they moved its lower end. Inside the last step Huxley's $3/(3\sigma - 1)$, which is below $4/(1+\sigma)$ from $\sigma = 7/9$ on, is below the step's exponent 2.25207 from $\sigma \approx 0.7774$ on, by up to $1.2 \cdot 10^{-2}$ at $131/168$; taking it there would not change the window, since the corners on that step do not bind. These slacks are the corner inequalities of `tools/linnik/route_limits.py`, evaluated on the curve with those steps changed.

The Halász–Montgomery branch

The top piece of the curve is the single step $101/50$ on $[5/6, 11/12]$ (`LinnikStep.charDensity_step`), and on it the window needs the Halász–Montgomery inequality (Section 3.6). Orthogonality alone would do with a smaller exponent near $11/12$: with $101/50$ kept on $[5/6, 7/8]$, where orthogonality allows any constant exponent up to 2.1356, and a constant exponent of at most 1.9595 on $[7/8, 11/12]$, every corner holds by orthogonality, and Huxley's estimate would give $3/(3 \cdot 7/8 - 1) + 1/50 = 24/13 + 1/50 \approx 1.866$ there, with the excess of the top piece. The library has no such step. Its steps of Huxley's estimate, `LinnikHuxStep.charDensity_gen`, bound the zeros that the integral detects by the fourth moment of the L-functions, and the condition that this asks of a step $[\sigma_-, \sigma_+]$ (`h3` of `LinnikHuxStep.Par.Good`) is $A(1-\sigma) \geq (201/100)(1/2 - \sigma) + 1/500 + 1 + 10^{-5}$ on the step, that is $A \geq 2.01 - 0.00299/(1 - \sigma_-)$; since the steps lie in $[3/4, 11/12]$, every one of them has $A > 2.01 - 12 \cdot 0.00299 > 1.974$. Huxley's exponents above $5/6$ are in the library only under Petrow and Young's Weyl bound for every character, taken as a hypothesis (`LinnikWeyl.PYWeyl`, in `LinnikStairs7.lean`), which empties that class of zeros and is not a theorem of the library. So with the density estimates that the library proves, the branch is needed.

The classical estimates in place of Chen, Gupta and Li's

Take g^* with its steps on $[5/7, 131/168]$ replaced by steps of length $1/1680$ with the classical exponents and the excess $1/2000$ of the library's steps: Ingham's [17] $3/(2 - \sigma_+) + 1/2000$ up to $3/4$ and Huxley's [25] $3/(3\sigma_- - 1) + 1/2000$ from $3/4$. The window is then set at $3/4$, where both exponents are $12/5$, and the closed forms of Section 3.5 become $u_1 \geq 16/(5L) - 1$ and $u_2 \leq 1 - 6/(5L)$, which meet at $L = 11/5$. The script `tools/linnik/density_payoff.py` (its variant `classical`, with `-deep`) finds a window admissible at L, checked at the corners in exact rationals, evaluates the

certificate's total on it on 272 cells a side under the rule at L, and lowers L in steps of 10^{-3} while the total stays below $0.999\mathcal{K}$ and the grid satisfies the conditions of Theorem 6.1, then tries one further step of $5 \cdot 10^{-4}$. It finds the window $[42950, 46385]/100000$ admissible at $L = 2239/1000 = 2.239$, and the certificate's total on it $0.94509\mathcal{K}$. The search stops there at the grid, not at the total: at 2.238 the window it finds, $[43014, 46361]/100000$, is too narrow for $20h \le v$ on 272 cells a side, so the script does not evaluate it, though `mirror8.py`'s total on that grid is $0.99209\mathcal{K}$; on 280 cells a side the window satisfies the grid conditions, but finer grids were not tried. So the sieve brings Huxley's 12/5 down to 2.239 at least, with Input 1 as the only input that is not refereed. The library has no steps of Ingham's or Huxley's estimate of length 1/1680 on $[5/7, 131/168]$ (its Ingham steps end at 5/7, and its Huxley steps, of length 1/280, start at 16/21); the parameters of `LinnikHuxStep.charDensity_gen` at this width were checked for the first of these steps only; and the certificate on this grid has not been checked by the kernel.

Below 11/5

At 11/5 the certificate's total is $0.9321382\mathcal{K}$ (Section 5); the kernel checks that it is below $0.9330704\mathcal{K}$. Of it, $0.7173\mathcal{K}$ comes from the 7289 candidate cells that are dropped, $0.6790\mathcal{K}$ of that from the 6896 of them that reach the line $\alpha_1 + \alpha_2 = 1 - U_1/D$, that is whose box's upper sum is at least $1 - U_1/D$; the decomposed cells cost $0.2149\mathcal{K}$. More depth can lower only the cost of the decomposed cells and of the 22 cells that are cheaper dropped: the other dropped cells, which cost $0.7151\mathcal{K}$, are dropped because the rule fails on their box or on a box of depth three below them (where the block test is the rule itself), and that does not depend on the depth. The two deepest levels matter for the rest: to depth six the total would be $1.55502\mathcal{K}$, since the cells of depth six that pass the block test would cost $0.669\mathcal{K}$ at their own bounds, against $0.049\mathcal{K}$ decomposed to depth eight. Without the bilinear test, that is with (R1) alone as the rule, 10974 candidate cells are decomposed in place of 15884, and the total would be $2.43064\mathcal{K}$. These counts and costs are printed by `tools/linnik/breakdown8.py`, and the totals without the deeper levels or without the bilinear test by its variants `depth6`, `depth4`, `lim` and `lim6`.

The cells that reach the line hold the integers that the Type II estimate cannot reach. Let $\varepsilon > 0$, and let $n = p_1 p_2 m \le 2x$ be an integer that such a cell counts, with $\alpha_1 + \alpha_2 \ge 1 - U_1/D + \varepsilon$ and α_1 and α_2 outside $\mathcal{U}$. For large x, no product of some of its prime factors has its exponent in $\mathcal{U}$, whatever the factorisation of m: a product with p_1 and another prime has

exponent at least $\alpha_1 + \alpha_2 > 1 - U_1/D$, since the prime factors of m are at least p_2; one without p_1 that omits p_2 or a prime of m has exponent at most $1 - \alpha_1 - \alpha_2 + o(1) < U_1/D$; and the others are p_1 and $p_2 m$, of exponents α_1, outside $\mathcal{U}$, and $1 - \alpha_1 + o(1) > 1/2$. In the decomposition such integers can be counted only by Type I estimates, and $p_1 p_2$ is beyond the Type I level X': the bilinear estimate of Lemma 4.2 reaches them where the bilinear test passes. Of the 6896 dropped cells that reach the line, 1725 fail the bilinear test themselves, 5162 pass it but have a box of depth three below them on which the rule fails, and 9 are cheaper dropped.

With the window chosen afresh, the method goes a little below 11/5. With the same curve, rule, depth and bounds, the window's corners checked in exact rationals and the grid satisfying the conditions of Theorem 6.1, the certificate's total is $0.96047\mathcal{K}$ at $L = 2.19925$, on the window $[43516, 46942]/100000$ and 272 cells a side; $0.96729\mathcal{K}$ at 2.199, on $[43538, 46935]/100000$ and 276 cells; and $1.01027\mathcal{K}$ at 2.198, on $[43617, 46911]/100000$ and 284 cells (`mirror8.py` and `route_limits.py corners` on these windows). So the method reaches about 2.198; 11/5, about $2 \cdot 10^{-3}$ above that, is the round exponent that the paper proves. A narrower window needs a finer grid: the condition $20h \le v$ of Theorem 6.1 asks for $N \ge 20(1/2 - v)/v$, which with $4 \mid N$ is $N \ge 268$ on the window of Proposition 5.4. Going much lower would need a wider window, that is a better density exponent near $\sigma = 5/7$, or Type I information on longer products than the bilinear estimate gives. The half-plane $\mathrm{Re}\, s > 7/8$ of OpenAI's main preprint [9], in place of Input 1, would not help: the window condition (Definition 3.3) would be needed only for $\beta \le 7/8$, and the exponent of the kernel bound would become $b(\mu) = 3/8 + \mu/8 + \delta'(1 - \mu) + 2\varepsilon'$, which is no larger for $\mu \le 1$; neither change reaches the corners that set the window's ends, on the first step at 5/7.

On the density exponent, Chen, Gupta and Li note that the worst case for their estimate is $T = 1$ with no divisor q_1 of q of a suitable size, as for prime q, that the critical situation is then at $\sigma = 5/7$, and that improving the term which dominates there is hard ([12, Remark 12.3], quoted in Section 8).

References

[1] U. V. Linnik, *On the least prime in an arithmetic progression. I. The basic theorem*, Rec. Math. [Mat. Sbornik] N.S. 15(57) (1944), 139–178.

[2] U. V. Linnik, *On the least prime in an arithmetic progression. II. The Deuring–Heilbronn phenomenon*, Rec. Math. [Mat. Sbornik] N.S.

15(57) (1944), 347–368.

[3] D. R. Heath-Brown, *Zero-free regions for Dirichlet L-functions, and the least prime in an arithmetic progression*, Proc. London Math. Soc. (3) 64 (1992), 265–338.

[4] T. Xylouris, *On the least prime in an arithmetic progression and estimates for the zeros of Dirichlet L-functions*, Acta Arith. 150 (2011), 65–91.

[5] T. Xylouris, *Über die Nullstellen der Dirichletschen L-Funktionen und die kleinste Primzahl in einer arithmetischen Progression*, Bonner Math. Schriften 404, Bonn, 2011.

[6] T. Xylouris, *Linniks Konstante ist kleiner als 5*, Chebyshevskii Sb. 19 (2018), 80–94.

[7] Y. Lamzouri, X. Li and K. Soundararajan, *Conditional bounds for the least quadratic non-residue and related problems*, Math. Comp. 84 (2015), 2391–2412.

[8] Y. Lamzouri, X. Li and K. Soundararajan, *Corrigendum to "Conditional bounds for the least quadratic non-residue and related problems"*, Math. Comp. 86 (2017), 2551–2554 (it corrects Lemma 6.1 and Theorems 1.2 and 1.3 of [7]).

[9] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$, preprint, 30 September 2026, `https://github.com/openai/math`, entry 003, read at commit `adc7f12`. Not refereed.

[10] OpenAI, *The Quasi-Riemann Hypothesis*, preprint, 5 October 2026, which proves $\mathrm{Re}(s) > 11/12$ by a different route from [9]; `https://github.com/openai/math`, entry 003, file `preprints/The-Quasi-Riemann-Hypothesis-October-5-2026/build/paper2.tex`, read at commit `adc7f12`. Not refereed.

[11] OpenAI, *Catalog of papers with a formalized main result*, file `lean/formalization.yaml` (version v0.4) of the repository `openai/math`, a web page, `https://github.com/openai/math/blob/main/lean/formalization.yaml`, read 25 October 2026; with `lean/OAI/NumberTheory/DirichletL/Nonvanishing.lean` of the

same repository. It is OpenAI's own record; it marks its scope as partial progress and its review status as unchecked.

[12] B. Chen, V. Gupta and Y. C. Li, *Large Value Estimates for Dirichlet Polynomials with Characters and Zero Density of Dirichlet L-Functions*, arXiv:2507.08296v2 (preprint; v2 of 27 July 2026, v1 of 11 July 2025 was by B. Chen alone).

[13] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) 203 (2026), 623–675; arXiv:2405.20552.

[14] D. R. Heath-Brown, *A large values estimate for Dirichlet polynomials*, J. London Math. Soc. (2) 20 (1979), 8–18.

[15] B. Kerr, *Large values of Dirichlet polynomials and zero density estimates for the Riemann zeta function*, arXiv:1909.12075.

[16] M. Jutila, *Zero-density estimates for L-functions*, Acta Arith. 32 (1977), 55–62.

[17] A. E. Ingham, *On the estimation of $N(\sigma, T)$*, Quart. J. Math. Oxford Ser. 11 (1940), 291–292.

[18] H. L. Montgomery, *Topics in Multiplicative Number Theory*, Lecture Notes in Math. 227, Springer, Berlin, 1971.

[19] H. Davenport, *Multiplicative Number Theory*, 3rd ed., revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, New York, 2000.

[20] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.

[21] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications 53, American Mathematical Society, Providence, RI, 2004.

[22] J. Friedlander and H. Iwaniec, *Opera de Cribro*, American Mathematical Society Colloquium Publications 57, American Mathematical Society, Providence, RI, 2010.

[23] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Graduate Studies in Mathematics 163, American Mathematical Society, Providence, RI, 2015.

[24] M. N. Huxley, *On the difference between consecutive primes*, Invent. Math. 15 (1972), 164–170.

[25] M. N. Huxley, *Large values of Dirichlet polynomials, III*, Acta Arith. 26 (1975), 435–444.

[26] H. M. Bui and D. R. Heath-Brown, *A note on the fourth moment of Dirichlet L-functions*, Acta Arith. 141 (2010), 335–344; arXiv:0903.4008.

[27] G. Halász, *Über die Mittelwerte multiplikativer zahlentheoretischer Funktionen*, Acta Math. Acad. Sci. Hungar. 19 (1968), 365–403.

[28] H. L. Montgomery, *Mean and large values of Dirichlet polynomials*, Invent. Math. 8 (1969), 334–345.

[29] H. L. Montgomery and R. C. Vaughan, *Hilbert's inequality*, J. London Math. Soc. (2) 8 (1974), 73–82.

[30] P. X. Gallagher, *The large sieve*, Mathematika 14 (1967), 14–20.

[31] P. X. Gallagher, *A large sieve density estimate near* $\sigma = 1$, Invent. Math. 11 (1970), 329–339.

[32] T. Kubota, *On Automorphic Functions and the Reciprocity Law in a Number Field*, Lectures in Mathematics, Department of Mathematics, Kyoto University, No. 2, Kinokuniya, Tokyo, 1969.

[33] S. J. Patterson, *A cubic analogue of the theta series*, J. Reine Angew. Math. 296 (1977), 125–161; II, ibid., 217–220.

[34] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, Ann. of Math. (2) 200 (2024), 967–1057.

[35] D. R. Heath-Brown, *A mean value estimate for real character sums*, Acta Arith. 72 (1995), 235–275.

[36] L. Goldmakher and B. Louvel, *A quadratic large sieve inequality over number fields*, Math. Proc. Cambridge Philos. Soc. 154 (2013), 193–212.

[37] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

[38] G. Harman, *On the distribution of αp modulo one*, J. London Math. Soc. (2) 27 (1983), 9–18.

[39] G. Harman, *Prime-Detecting Sieves*, London Math. Soc. Monographs (New Series) 33, Princeton University Press, 2007.

[40] D. R. Heath-Brown and H. Iwaniec, *On the difference between consecutive primes*, Invent. Math. 55 (1979), 49–69.

[41] R. C. Baker, G. Harman and J. Pintz, *The difference between consecutive primes, II*, Proc. London Math. Soc. (3) 83 (2001), 532–562.

[42] R. Li, *The number of primes in short intervals and numerical calculations for Harman's sieve*, arXiv:2308.04458v8 (preprint, 16 October 2025); Theorems 9 and 11 are on p. 36.

[43] W. D. Banks and I. E. Shparlinski, *Bounds on short character sums and L-functions with characters to a powerful modulus*, J. Anal. Math. 139 (2019), 239–263.

[44] V. Brun, *Le crible d'Eratosthène et le théorème de Goldbach*, Videnskapsselskapets Skrifter I, Mat.-naturv. Klasse, Kristiania, 1920, no. 3.

[45] A. A. Buchstab, *Asymptotische Abschätzung einer allgemeinen zahlentheoretischen Funktion*, Rec. Math. [Mat. Sbornik] N.S. 2(44) (1937), 1239–1246.

[46] A. Y. Cheer and D. A. Goldston, *A differential delay equation arising from the sieve of Eratosthenes*, Math. Comp. 55 (1990), 129–141.

[47] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. 12699, Springer, 2021, 625–635.

[48] The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381.

[49] R. Comley, *number-theory*, the repository of the paper, its supplement and its Lean library, `https://github.com/dickie81/number-theory`. The paper and its supplement describe the library at the commit that contains them; the earlier version of the library, at the exponent

278/125, is at commit `8ba053b`: `https://github.com/dickie81/number-theory/tree/8ba053b75eb56a65dcfd1cf986ecb3d58c6945bb`.

[50] AIProofWatch, record of OpenAI's *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$, `https://aiproofwatch.com/updates/openai-math-003` (assessment reviewed 25 October 2026; reports an external Lean check of the Riemann zeta specialisation only).

[51] A. Kontorovich, T. Tao and contributors, *PrimeNumberTheoremAnd*, a Lean project, `https://github.com/AlexKontorovich/PrimeNumberTheoremAnd`, at commit `650d312`.

[52] Anthropic, *formal-math*, a Lean repository, `https://github.com/anthropics/formal-math`, at commit `fbdc36b`.